

zkBugs Evaluation Results

Table 1: Tool Performance Summary

Tool	TP	FN	Timeout	Failure	Total	Median Time (s)
circomspect	0	70	0	0	70	0.03
circom_civer	12	11	22	25	70	0.88
picus	10	35	19	6	70	2.62
ecneproject	25	38	6	1	70	22.21
zkfuzz	25	16	25	4	70	0.97
circom_auditor	60	3	2	2	67	599.81
TOTAL	132	173	74	38	417	16.06

Table 2: Execution Time Statistics

Tool	Total	Min (s)	Max (s)	Median (s)	Mean (s)	NR-Timeout
circomspect	69	0.01	0.33	0.03	0.07	0
circom_civer	23	0.04	502.33	0.88	86.00	22
picus	51	1.08	494.72	2.62	77.59	19
ecneproject	63	12.93	148.34	22.21	30.10	6
zkfuzz	41	0.01	531.69	0.97	40.41	25
circom_auditor	66	174.20	1714.22	599.81	693.02	2

Table 3: Bug-Tool Matrix (by Bug ID)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz	circom_auditor
1	FN	FN	FN	FN	FN	FN
2	FN*	Timeout	FN	Timeout	Timeout	Timeout
3	FN	Timeout	Timeout	TP*	TP*	TP
4	FN	FN	FN	FN	TP*	TP
5	FN	FN	Failure	FN	TP*	TP
6	FN	Failure	FN	FN	Failure	TP
7	FN	Failure	Timeout	TP*	Failure	TP
8	FN*	Failure	FN	FN	Failure	TP*
9	FN	TP	TP*	TP*	TP*	TP
10	FN*	Timeout	Timeout	TP*	TP*	TP*
11	FN	TP	TP*	TP*	TP*	TP
12	FN	Timeout	FN	FN	Timeout	TP
13	FN	Timeout	FN	FN	Timeout	TP*
14	FN	FN	Timeout	FN	TP*	TP
15	FN	FN	Timeout	FN*	Timeout	Undecided*
16	FN	FN	Timeout	FN	TP*	TP*
17	FN	Timeout	Timeout	TP*	FN	TP*
18	FN	FN	FN*	FN*	Timeout	TP*
19	FN	FN	FN	FN	Timeout	TP
20	FN	Timeout	Timeout	Timeout	FN	TP
21	FN*	Timeout	FN	Timeout	Failure	Undecided*
22	FN*	Timeout	Timeout	Timeout	FN	FN*
23	FN*	Timeout	Timeout	FN	Timeout	TP*
24	FN*	TP	TP*	TP*	TP*	Timeout
25	FN	Failure	FN	FN	FN	Failure
26	FN*	Failure	FN*	FN*	Timeout	TP*
27	FN*	Failure	FN	FN	FN	Failure
28	FN	Failure	FN	FN	Timeout	TP*
29	FN*	Failure	FN*	FN*	Timeout	TP*
30	FN	TP	TP*	TP*	TP*	TP*
31	FN*	Failure	FN*	FN*	TP*	TP*
32	FN*	FN	FN*	FN*	FN*	TP*
33	FN	Failure	FN*	Timeout	FN	TP*
34	FN	Timeout	Timeout	TP*	Timeout	TP*
35	FN	Timeout	FN	FN	Timeout	TP
36	FN	FN	FN	FN	Timeout	TP*
37	FN	Failure	FN	Timeout	FN	TP
38	FN	Failure	FN	FN	Timeout	TP
39	FN	Timeout	FN*	FN*	Timeout	TP*

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Bug ID	circospect	circom_civer	picus	ecneproject	zkfuzz	circom_auditor
40	FN*	FN	FN	FN	Timeout	TP
41	FN*	Timeout	Timeout	TP*	TP*	TP
42	FN	Timeout	Timeout	FN	Timeout	TP*
43	FN	TP*	Failure	TP*	TP*	TP*
44	FN	TP*	Failure	TP*	TP*	TP
45	FN	TP*	Failure	TP*	TP*	TP
46	FN	TP	TP*	TP*	TP*	TP*
47	FN	TP	TP*	TP*	TP*	TP*
48	FN	TP	TP*	TP*	TP*	TP
49	FN	Timeout	TP*	TP*	TP*	TP*
50	FN*	Failure	FN	FN	Timeout	TP
51	FN	Failure	FN	FN	FN	TP*
52	FN	Failure	FN	FN	FN	TP
53	FN*	Timeout	Timeout	TP*	FN	TP
54	FN	Failure	FN	FN	TP*	TP
55	FN	Failure	FN	Failure	FN	TP*
56	FN	Timeout	FN	FN	Timeout	TP
57	FN	Timeout	Timeout	TP*	Timeout	TP*
58	FN	Timeout	Timeout	TP*	TP*	TP
59	FN*	TP	TP*	TP*	TP*	TP
60	FN*	Timeout	Failure	TP*	TP*	TP*
61	FN	Failure	Timeout	TP*	Timeout	TP
62	FN*	Failure	FN*	FN*	FN*	TP*
63	FN*	Failure	FN*	FN*	FN*	TP*
64	FN	Failure	FN	FN	FN	TP
65	FN*	Failure	Failure	FN	Timeout	TP
66	FN	Failure	Timeout	TP*	Timeout	TP*
67	FN	Failure	FN	FN	Timeout	FN
68	FN	Timeout	Timeout	FN	Timeout	TP*
69	FN*	Failure	FN	FN	TP*	Undecided
70	FN*	TP	TP*	TP*	TP*	TP*

* = *manually analyzed*

Table 4: Bug ID to Name Mapping

ID	Bug Name
1	daira_hopwood_darkforest_v0_3_missing_bit_length_check
2	hashcloak_data_are_not_fully_verified_during_state_update
3	kobi_gurkan_mimc_hash_assigned_but_not_constrained
4	koukyosyumei_checkdepth_comparator_overflow
5	koukyosyumei_clamp_comparator_overflow
6	koukyosyumei_emitifinrange_lessthan_overflow
7	koukyosyumei_extractstringfrompoint_shiftedfirstbyte_unconstrained
8	koukyosyumei_spend_missing_range_check
9	koukyosyumei_subtract_unconstrained_multiplier
10	koukyosyumei_unconstrained_base64_decoded_len
11	koukyosyumei_under_constrained_date_encoder
12	leastauthority_Circuit_Does_Not_Check_the_ERC_20_Sum_Correctly_
13	leastauthority_previously_correct_ownership_proof_disabled_via_code_changes
14	matterlabs_email_spoofing_via_manipulated_from_header
15	matterlabs_emailauth_fails_with_overlapping_invitation_code_regex
16	matterlabs_fromaddrregex_allows_email_address_spoofing
17	matterlabs_underconstrained_fpmul_circuit
18	openzeppelin_mismatched_base64url_decoding_may_break_completeness
19	openzeppelin_non_determinism_of_some_inputs
20	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
21	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
22	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
23	trailofbits_unsafe_use_of_num2bits_in_multiple_circuits
24	veridise_arrayxor_is_under_constrained
25	veridise_babyjubjub_suborder_constraints_not_applied_correctly
26	veridise_blacklist_states_not_representable_in_field
27	veridise_bypassing_internal_transfer_limits_via_swaps
28	veridise_creditnoteproofs_can_be_stolen
29	veridise_data_escrow_encrypted_message_wrong_input
30	veridise_decoder_accepting_bogus_output_signal
31	veridise_fortxreward_abstracted_away_in_reward_calc
32	veridise_incorrect_initialization_in_membership_circuits
33	veridise_kyt_signature_verification_fails_nonzero_hash
34	veridise_missing_range_checks_in_bigmod
35	veridise_missing_range_checks_on_comparison_circuits
36	veridise_no_zero_value_validation
37	veridise_nullifier_verification_can_be_disabled
38	veridise_ownershipproofs_could_identify_issuers
39	veridise_private_information_leakage

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ID	Bug Name
40	veridise_spammers_may_slash_themselves
41	veridise_template_CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
42	veridise_underconstrained_circuit_allows_invalid_comparison
43	veridise_underconstrained_outputs_in_bitElementMulAny
44	veridise_underconstrained_outputs_in_window4
45	veridise_underconstrained_outputs_in_windowmulfix
46	veridise_underconstrained_points_in_edwards2Montgomery
47	veridise_underconstrained_points_in_montgomery2Edwards
48	veridise_underconstrained_points_in_montgomeryAdd
49	veridise_underconstrained_points_in_montgomeryDouble
50	veridise_unsafe_num2bits_254_blacklist_leaf
51	veridise_zaccount_renewal_kyc_expiry_not_validated
52	veridise_zaccount_renewal_multiple_nullifiers_same_utxo
53	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
54	veridise_zone_id_inclusion_prover_bypass
55	veridise_zone_limits_bypass_zswapv1
56	yacademy_Knowledge_of_any_member_signature_allow_to_generate_proof_of_membership
57	yacademy_input_signal_s_is_not_constrained_in_eff_ecdsa_circom
58	yacademy_under_constrained_circuits_compromising_the_soundness_of_the_system
59	zksecurity_Unsound_Addition_Gadget
60	zksecurity_Unsound_XOR_gadget
61	zksecurity_an_attacker_can_craft_a_fake_non_inclusion_proof_for_a_given_key_due_to_an_aliasing_bug_in_the_smt_verifier
62	zksecurity_big_integer_zero_check_is_not_sound
63	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
64	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
65	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrary_L
66	zksecurity_missing_byte_range_checks_allows_packed_data_pollution
67	zksecurity_second_pre_image_attacks_on_packbytesandposeidon_may_be_used_to_register_arbitrary_passports_and_dsc_certificates
68	zksecurity_sha256_templates_return_zero_on_arbitrary_inputs
69	zksecurity_the_registration_and_disclosure_circuits_lack_range_checks_for_the_input_indices
70	zksecurity_unsound_left_rotation

Table 5: Execution Times per Bug (by Bug ID, in seconds)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz	circom_auditor
1	0.02	0.15	1.85	22.80	89.63	459.80
2	0.30	—	333.50	—	—	—
3	0.01	—	—	20.73	0.04	564.41
4	0.01	0.14	1.19	13.40	0.03	758.19
5	0.01	0.13	238.67	23.09	0.02	587.87
6	0.01	—	1.48	17.50	—	543.22
7	0.01	—	—	18.17	—	459.96
8	0.02	—	2.61	27.04	—	515.05
9	0.01	0.07	2.31	25.55	0.02	174.20
10	0.03	—	—	26.94	0.13	516.79
11	0.01	0.29	2.62	23.19	2.63	294.06
12	0.18	—	10.77	48.44	—	910.63
13	0.16	—	4.36	32.31	—	501.40
14	0.02	66.86	—	20.95	1.38	701.63
15	0.02	249.13	—	18.93	—	863.49
16	0.02	49.64	—	16.06	0.97	953.24
17	0.02	507.20	—	23.56	15.01	481.92
18	0.04	0.88	446.72	23.64	—	375.18
19	0.02	64.33	21.59	76.66	—	596.49
20	0.06	—	—	—	42.87	1204.22
21	0.18	—	2.07	—	2.45	1351.33
22	0.23	—	—	—	9.71	1493.40
23	0.01	500.26	—	19.82	—	366.70
24	0.06	0.13	1.32	15.82	0.12	—
25	0.10	—	3.48	29.35	6.79	772.93
26	0.13	—	2.94	22.41	—	976.67
27	0.09	—	3.53	28.22	461.40	571.76
28	0.02	—	3.85	22.89	—	685.75
29	0.26	—	17.80	91.09	—	911.23
30	0.01	0.04	1.35	18.08	0.09	338.90
31	0.10	—	1.87	21.03	0.10	713.22
32	0.04	0.06	1.98	27.61	16.65	208.57
33	0.32	—	444.92	—	39.30	1325.13
34	0.03	—	—	27.53	—	804.74
35	0.01	504.09	1.16	12.93	—	483.12
36	0.19	13.99	4.85	34.13	—	477.50
37	0.29	—	439.58	—	40.47	1198.49
38	0.07	—	1.11	13.70	—	371.79
39	0.11	—	7.91	44.08	—	692.12

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz	circom_auditor
40	0.10	4.02	1.54	15.08	—	268.78
41	0.24	501.91	—	34.89	0.86	814.43
42	0.03	553.48	—	18.06	—	648.92
43	0.02	502.19	81.75	16.91	0.05	554.35
44	0.04	502.33	338.44	20.02	3.22	1122.25
45	0.02	502.30	338.24	20.33	1.69	647.76
46	0.01	0.09	1.59	18.91	0.41	314.11
47	0.01	0.04	1.34	16.19	0.08	394.81
48	0.01	2.06	1.43	16.31	0.01	318.17
49	0.01	500.10	1.48	16.74	0.01	381.54
50	0.15	—	3.45	25.29	—	663.55
51	0.26	—	55.23	141.61	86.32	1045.03
52	0.24	—	56.28	142.66	99.27	1021.24
53	0.06	500.16	—	22.96	531.69	1058.52
54	0.04	—	1.53	18.87	0.16	603.13
55	0.33	—	454.02	328.75	41.22	1714.22
56	0.05	—	3.88	33.87	—	904.13
57	0.04	—	—	24.90	—	823.43
58	0.03	—	—	25.84	0.22	1051.11
59	0.01	0.07	1.69	22.30	0.02	261.26
60	0.01	500.26	494.72	22.21	0.03	272.69
61	0.01	—	—	19.49	—	1347.74
62	0.01	—	1.08	14.07	17.04	572.08
63	0.01	—	1.08	15.92	71.24	500.12
64	0.03	—	1.47	17.31	74.86	687.82
65	0.01	—	104.05	18.91	—	534.53
66	0.03	—	—	27.95	—	1381.39
67	0.02	—	1.60	13.21	—	359.03
68	0.04	—	—	148.34	—	842.45
69	0.01	—	1.47	17.48	0.02	480.49
70	—	19.04	2.53	22.17	0.84	289.59